

Snake Game Learning Summary

What You Built

Built a Pygame window, game loop, keyboard controls, snake movement, random food, snake growth, wall collision, self collision, and score logic.

Important Variables

width, height, screen, snake, snake_size, speed, dx, dy, food_x, food_y, score, running.

Important Modules

pygame, pygame.display, pygame.event, pygame.time, pygame.draw, pygame.font, random.

Functions & Methods

pygame.init(), set_mode(), set_caption(), event.get(), draw.rect(), display.update(), quit(), Clock(), tick(), SysFont(), render(), blit(), random.randrange(), snake.insert(), snake.pop().

Algorithms

Movement: get head -> create new head -> insert -> pop. Growth: skip pop when food is eaten. Food collision: compare head and food. Wall collision: check four boundaries. Self collision: compare head with snake[1:]. Score: increment when food is eaten. Text: SysFont -> render -> blit -> update.

Programming Concepts

Lists, nested lists, indexing, tuples vs lists, loops, conditions, modules, objects, methods, game loop structure, problem solving.

Next Session

Finish displaying the score, build a Game Over screen, add restart functionality, and polish the game.